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Top 100 Microsoft Solution Architect Interview Questions



Microsoft Solution Architect is a role within the Microsoft Certified: Azure Solutions Architect Expert certification. It involves designing and implementing solutions that run on Microsoft Azure, a cloud computing platform. Microsoft Solution Architects are responsible for understanding business requirements and technical specifications and creating an appropriate solution that meets those needs.

But, for an interview, you need to focus both on your interpersonal and solution architect skills. To help you ou, in this blog we will be discussing the top Microsoft Solution Architect questions that will help you in passing the interview.

Advanced Sample Questions

What are the benefits of using Azure over other cloud platforms?

Azure offers a wide range of benefits, including scalability, cost-effectiveness, security, reliability, and flexibility. It also has a large ecosystem of tools and services that can be easily integrated with other Microsoft products.

How do you design a highly available solution on Azure?

To design a highly available solution on Azure, you need to consider factors such as the availability of the underlying infrastructure, the application architecture, data replication, load balancing, and failover mechanisms. You can use Azure features such as Availability Zones, Load Balancer, and Traffic Manager to ensure high availability.

How do you ensure data security on Azure?

You can ensure data security on Azure by using features such as Azure Security Center, Azure Key Vault, Azure Active Directory, and Azure Multi-Factor Authentication. You should also implement strong access controls, data encryption, and monitoring and auditing processes.

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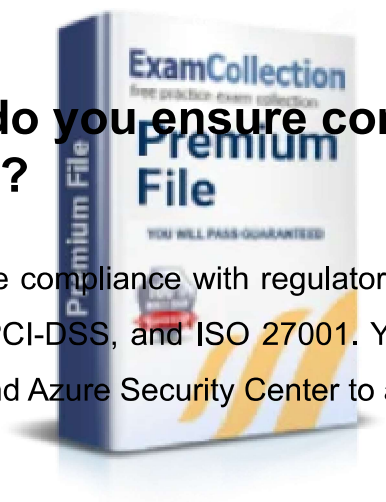
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How do you handle disaster recovery on Azure?

You can handle disaster recovery on Azure by using features such as Azure Site Recovery, Azure Backup, and Azure Virtual Machines. These tools provide backup and replication capabilities that allow you to quickly recover from disasters.



How do you ensure compliance with regulatory requirements on Azure?

To ensure compliance with regulatory requirements on Azure, you should follow industry standards such as HIPAA, PCI-DSS, and ISO 27001. You can use Azure features such as Azure Compliance Manager, Azure Policy, and Azure Security Center to assess compliance and enforce policies.

How do you design a scalable solution on Azure?

To design a scalable solution on Azure, you need to consider factors such as resource utilization, load balancing, auto-scaling, and caching. You can use Azure features such as Azure Autoscale, Azure Load Balancer, and Azure Cache for Redis to ensure scalability.

How do you secure Azure resources?

To secure Azure resources, you should implement access controls, use strong authentication mechanisms, encrypt data, and monitor and audit activity. You can use features such as Azure Security Center, Azure Key Vault, and Azure Active Directory to enhance security.

How do you ensure the interoperability of applications and systems in a complex environment?

To ensure the interoperability of applications and systems in a complex environment, a Microsoft Solution Architect should consider the following:

1. Use standardized protocols and formats: Using standardized protocols and formats such as REST, SOAP, and JSON can help ensure that systems can communicate with each other seamlessly.
2. Leverage API gateways: An API gateway can act as a central point for managing and routing requests between different systems. By implementing an API gateway, the Solution Architect can ensure that systems can communicate with each other, regardless of their underlying technology.
3. Adopt microservices architecture: A microservices architecture can help break down complex systems into smaller, more manageable services. This approach can help improve interoperability by enabling different services to be developed independently of each other.
4. Implement service-oriented architecture (SOA): SOA is an architectural style that emphasizes the use of loosely coupled services. By implementing SOA, a Solution Architect can create a system where services can communicate with each other seamlessly, even if they are developed using different technologies.
5. Use messaging queues: Messaging queues can be used to enable asynchronous communication between systems. This approach can help ensure that systems can communicate with each other, even if one system is temporarily unavailable.
6. Ensure data consistency: To ensure interoperability, the Solution Architect must ensure that all systems are using the same data model and that data is consistent across different systems.

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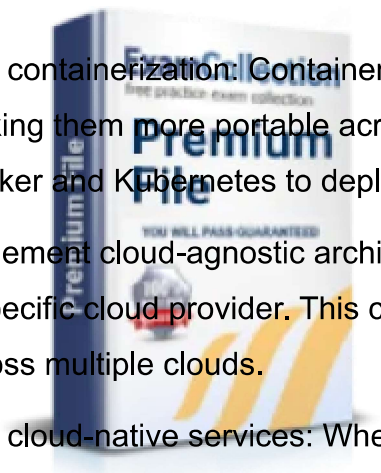
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How do you ensure the portability of applications and data in a multi-cloud environment?

To ensure the portability of applications and data in a multi-cloud environment, a Microsoft Solution Architect should consider the following:



1. Use containerization: Containerization allows applications to be packaged with all their dependencies, making them more portable across different environments. Solutions Architects can use tools like Docker and Kubernetes to deploy and manage containers across different clouds.
2. Implement cloud-agnostic architectures: Solutions Architects should design systems that are not tied to a specific cloud provider. This can be achieved by using open-source tools and technologies that work across multiple clouds.
3. Use cloud-native services: When designing solutions, Solutions Architects should consider using cloud-native services that are available across different clouds. For example, using services like AWS Lambda, Azure Functions, or Google Cloud Functions can make it easier to move applications between clouds.
4. Implement a data management strategy: To ensure data portability, Solutions Architects must consider how data is stored, accessed, and moved between different clouds. Solutions Architects should consider using open standards for data storage and integration, such as SQL and REST.
5. Implement a multi-cloud management platform: A multi-cloud management platform can provide a single interface to manage multiple clouds, making it easier to deploy and manage applications and data across different environments.
6. Ensure security and compliance: Solutions Architects must ensure that their solutions are secure and compliant across all the clouds they use. They should consider using cloud-native security tools, as well as tools that can provide compliance across multiple clouds.

Can you explain your experience with designing and implementing microservices architectures?

A microservices architecture is an approach to software design and development that involves breaking down an application into smaller, independently deployable services that communicate with each other over a network. Each microservice is designed to perform a specific business function and can be developed, deployed, and scaled independently of the others.

The benefits of a microservices architecture include increased flexibility, scalability, and resilience, as well as the ability to use different technologies and programming languages for different services. However, designing and implementing a microservices architecture can also be complex and requires careful planning and consideration of various factors, such as service boundaries, data management, communication protocols, and deployment strategies.

Some best practices for designing and implementing a microservices architecture include using a domain-driven design approach to identify service boundaries, ensuring loose coupling between services, adopting standard communication protocols such as REST or gRPC, implementing automated testing and deployment processes, and using containerization technologies such as Docker and Kubernetes for deployment and management.

Overall, designing and implementing a microservices architecture can be a challenging but rewarding process that requires careful consideration of various factors and a commitment to best practices and continuous improvement.

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What is your experience with implementing identity and access management solutions in the cloud?

In the context of the cloud, IAM solutions are used to manage access to cloud-based resources, such as virtual machines, storage, and applications. Cloud IAM solutions typically use a combination of authentication mechanisms, such as passwords, multi-factor authentication, and single sign-on, and authorization mechanisms, such as role-based access control and attribute-based access control.

When implementing IAM solutions in the cloud, there are several key considerations to keep in mind. These include:

1. Choosing the right IAM provider: There are many IAM providers in the market, and it's important to choose one that meets your organization's needs in terms of features, scalability, and security.
2. Defining roles and permissions: Before implementing an IAM solution, it's important to define roles and permissions for users and resources to ensure that access is granted only to authorized users.
3. Enforcing access policies: Access policies should be defined and enforced to ensure that users can only access resources that they are authorized to use.
4. Monitoring access: IAM solutions should be configured to log user access to resources to detect unauthorized access attempts and provide audit trails for compliance purposes.

Overall, implementing IAM solutions in the cloud can help organizations manage access to their cloud-based resources in a secure and scalable way. However, it's important to carefully consider the various factors involved in implementing IAM solutions and to follow best practices to ensure that access is granted only to authorized users.

Can you explain how you would optimize costs in a cloud environment?

Optimizing costs in a cloud environment is important for organizations that want to maximize their return on investment and reduce unnecessary expenses. Here are some strategies that can be used to optimize costs in a cloud environment:

1. Right-size resources: One of the biggest advantages of the cloud is the ability to scale resources up and down as needed. By choosing the right size for your virtual machines, storage, and other cloud resources, you can avoid paying for more resources than you need.
2. Use reserved instances: Reserved instances are a way to save money by committing to use a specific instance type for a set period of time. This can result in significant cost savings compared to on-demand instances.
3. Leverage auto-scaling: Auto-scaling can be used to automatically adjust the number of resources in use based on demand. This can help to avoid over-provisioning resources and reduce costs.
4. Optimize storage usage: By using tiered storage and deleting unused resources, you can reduce storage costs.
5. Use spot instances: Spot instances are a way to bid on unused computing capacity, which can be significantly cheaper than on-demand instances. This approach requires some flexibility, as the capacity may be reclaimed at any time, but can be a cost-effective option for certain workloads.
6. Monitor and analyze usage: By monitoring cloud usage and analyzing trends, you can identify areas where resources are being over-provisioned or under-utilized. This can help you to make more informed decisions about how to optimize costs.
7. Choose the right pricing model: Different cloud providers offer different pricing models, such as pay-as-you-go or upfront payment. By choosing the right pricing model for your organization's needs, you can

reduce costs and avoid unnecessary expenses.

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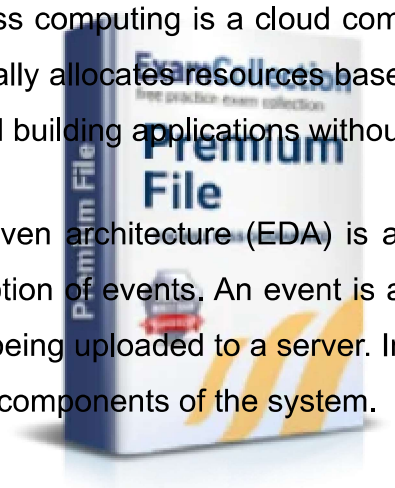
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What is your experience with serverless computing and event-driven architectures?



Serverless computing is a cloud computing model where the cloud provider manages the infrastructure and dynamically allocates resources based on the application's needs. This allows developers to focus on writing code and building applications without having to worry about managing servers or scaling infrastructure.

Event-driven architecture (EDA) is a software architecture that emphasizes the production, detection, and consumption of events. An event is a signal that something has happened, such as a user clicking a button or a file being uploaded to a server. In an EDA, events trigger actions or responses, which can be handled by different components of the system.

Serverless computing and event-driven architectures are often used together to build scalable and responsive applications. In a serverless architecture, individual functions can be triggered by events, allowing for a highly responsive system that can handle varying loads. This also allows for the creation of event-driven workflows, where different functions are executed in response to specific events.

How do you ensure performance and scalability in a distributed system?

Ensuring performance and scalability in a distributed system can be a complex and challenging task, but there are several strategies that can help. Here are some of the key considerations when designing and implementing a distributed system to ensure performance and scalability:

1. Partitioning: Partitioning involves dividing data and processing across multiple nodes in the system. This allows for better load balancing and can improve performance and scalability. There are several types of partitioning strategies, including horizontal partitioning (sharding), vertical partitioning (splitting tables by columns), and functional partitioning (separating functionality based on different nodes).
2. Caching: Caching involves storing frequently accessed data in memory or on a separate cache layer. This can reduce the load on the system and improve performance by reducing the number of requests to the database or other data sources.
3. Load balancing: Load balancing involves distributing traffic across multiple servers or nodes. This can help to prevent overloading of individual nodes and ensure that requests are processed efficiently.
4. Replication: Replication involves copying data to multiple nodes in the system. This can improve performance and availability by allowing for faster access to data and reducing the risk of data loss.
5. Asynchronous communication: Asynchronous communication allows for non-blocking communication between nodes in the system. This can improve performance and scalability by allowing nodes to continue processing requests while waiting for a response.
6. Monitoring and analysis: Monitoring and analyzing system performance is critical for identifying bottlenecks and areas for improvement. This can involve using tools such as performance metrics, logging, and tracing to identify issues and optimize system performance.
7. Horizontal scaling: Horizontal scaling involves adding more nodes to the system as needed. This can be done manually or automatically based on system load, and can help to ensure that the system can handle increasing levels of traffic and data processing.

Can you explain your experience with DevOps practices and tools like Jenkins, Ansible, and Terraform?

DevOps is a set of practices and tools that combine development and operations to improve the speed, quality, and reliability of software delivery. It involves a culture shift that promotes collaboration and communication between development and operations teams, as well as the use of automation and monitoring tools to streamline the software delivery process.

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Jenkins is an open-source automation server that is used to automate software development processes such as building, testing, and deploying software. It provides a wide range of plugins that can be used to automate tasks and integrate with other tools and services.

Ansible is an open-source IT automation tool that is used to automate tasks such as configuration management, application deployment, and infrastructure orchestration. It uses a simple, human-readable language to define tasks and can be used to manage systems across multiple platforms.

Terraform is an open-source tool for building, changing, and versioning infrastructure. It allows developers to define infrastructure as code, which can be versioned, reviewed, and tested just like application code. Terraform supports a wide range of cloud providers and can be used to manage infrastructure across multiple environments.

How do you ensure data privacy and protection in a cloud environment?

Ensuring data privacy and protection in a cloud environment is a critical responsibility for any organization that processes sensitive data. Here are some best practices to ensure data privacy and protection in a cloud environment:

1. Choose a secure cloud provider: Choose a cloud provider that has a strong security and privacy track record, and that meets industry and regulatory compliance standards. Look for providers that offer data encryption at rest and in transit, strong access controls, and advanced security features.
2. Encrypt data: Encrypt all sensitive data before it is stored in the cloud. This will ensure that even if data is compromised, it cannot be read without the decryption key. You can use a variety of encryption methods such as SSL/TLS, AES, or RSA.
3. Manage access controls: Manage access controls to ensure that only authorized personnel have access to sensitive data. This includes implementing strong passwords, multi-factor authentication, and role-based access controls.
4. Implement data backup and disaster recovery: Implement a backup and disaster recovery plan to ensure that data can be recovered in the event of a data breach or other disaster. This includes regular backups and testing of data recovery processes.
5. Monitor and audit access: Monitor and audit all access to sensitive data in the cloud environment. This includes logging and tracking all user activity, monitoring for suspicious behavior, and implementing intrusion detection and prevention systems.
6. Stay up-to-date on security threats: Stay up-to-date on the latest security threats and vulnerabilities, and implement security patches and updates as soon as they are released.

Can you explain your experience with implementing and managing hybrid cloud architectures?

Implementing and managing hybrid cloud architectures requires careful planning and consideration of factors such as data security, network connectivity, and workload placement. Some key considerations include:

1. Data security: Protecting sensitive data is critical in a hybrid cloud environment. Organizations need to ensure that data is encrypted at rest and in transit, and that access controls are in place to prevent unauthorized access.

2. Network connectivity: To ensure seamless operation between public and private cloud environments, organizations need to ensure that they have adequate network connectivity and bandwidth. This may involve using virtual private networks (VPNs) or other technologies to securely connect cloud environments.

3. Workload placement: To optimize performance and cost-effectiveness, organizations need to carefully consider which workloads are best suited for public cloud versus private cloud or on-premises infrastructure. This may involve analyzing workload requirements and performance characteristics, as well as assessing cost and compliance considerations.

4. Integration: To ensure seamless operation between public and private cloud environments, organizations need to integrate different systems and applications using APIs and other integration technologies.

5. Management and monitoring: To ensure optimal performance and availability, organizations need to manage and monitor their hybrid cloud environments using tools and technologies that provide visibility into performance, usage, and security.

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Basic Sample Questions

Can you name the principal segments of the Azure platform?

There are three principal segments in Azure:

1. Windows Azure Compute

This segment provides code that a hosting environment manages. Moreover, it consists of three roles which are Web Role, Worker Role, and VM Role.

2. Windows Azure Storage

This provides storage solutions using the services like Queue, Tables, Blobs, and Windows Azure Drives (VHD).

3. Windows Azure AppFabric

This consists of services like Service bus, Access, Caching, Integration, and Composite.

When an issue is said to be break-fix in Azure?

Break-Fix situation refers to the technical fault that arises when the functions designed for supporting the performance of technology fail to achieve their core implementation.

What do you understand by the Azure deployments slot?

Deployment slots located under the Azure Web App Service. They are basically of two types, Production slot, and Staging slot. Where the production slot refers to the default one that is used for running applications. And the staging slots help in testing the application usability before promoting to the production slot.

Explain the ways for managing the session state in Azure?

For managing the sessions state you can use SQL Azure, Windows Azure Caching, and Azure Table.

Explain the process for communicating with two Virtual Networks?

For creating communication between two Virtual Network there is a requirement for firstly, creating a Gateway subnet. The gateway subnet is configured while defining the range of the Virtual network. Further, it uses the IP addresses for specifying the quantity of subnet to be contained.

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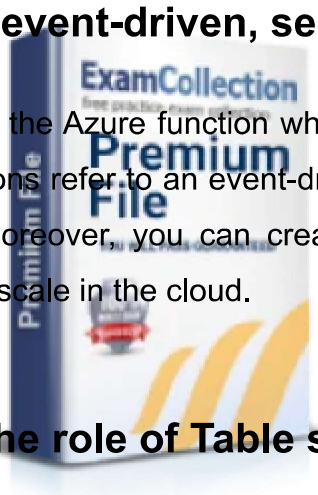
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Name the Azure service which can help in speeding up the app development using an event-driven, serverless architecture.

You can use the Azure function which will help in developing more efficiently with Functions. That is to say, Azure functions refer to an event-driven serverless compute platform used for solving complex orchestration problems. Moreover, you can create and debug locally without any need for setting up, deploying, and operating at scale in the cloud.

What is the role of Table storage in Azure?

Azure Table storage is use for storing non-relational structured data in the cloud by providing a key/attribute store with a strategic design. This stores flexible datasets like

- Firstly, user data for web applications address books
- Secondly, device information
- Lastly, types of metadata.
    - *Further, it has the* capability of *storing large amounts of structured data.*

Can you provide some of the uses of Azure table storage?

The Common uses of Table storage include:

- Firstly, storing TBs of structured data having ability to serve web-scale applications
- Secondly, storing datasets that don’t need complex joins, foreign keys, or stored procedures and can be denormalize for fast access
- Then, using a clustered index for quickly querying data
- Lastly, using the OData protocol and LINQ queries with WCF Data Service .NET Libraries for accessing data.

What are the major uses of Azure Blob Storage?

This helps in:

- Firstly, serving images or documents directly to a browser.
- Secondly, storing files for distributed access.
- Thirdly, streaming video and audio.
- Then, writing to log files.
- Lastly, storing data for backup and restore disaster recovery, and archiving.

Define the following in Blob Storage.

1. Storage Account

A storage account is for providing a unique namespace in Azure for your data. Every object stored in Azure Storage has an address that includes your unique account name. Further, the combination of the account

name and the Azure Storage blob endpoint creates the base address for the objects in your storage account.

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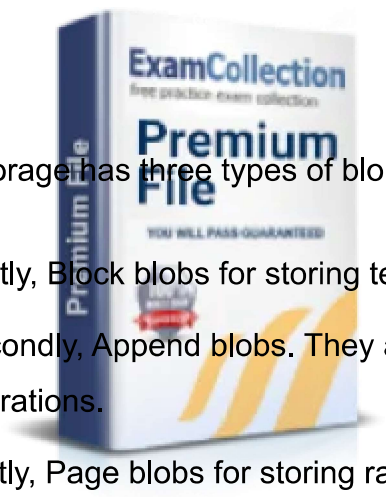
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2. Containers

A container is for organizing a set of blobs to a directory in a file system. There can be an unlimited number of containers in a storage account and a container can store an unlimited number of blobs.

3. Blobs



Azure Storage has three types of blobs:

- Firstly, Block blobs for storing text and binary data.
- Secondly, Append blobs. They are built from blocks like block blobs but they perform append operations.
- Lastly, Page blobs for storing random access files up to 8 TiB in size.

Explain Azure Active Directory (AD) service?

Azure Active Directory (Azure AD) refers to a multi-tenant cloud-based identity and directory management service which is a mixture of core directory services, application access management, and identity protection.

What is Azure Load Balancer?

Azure Load Balancer runs at layer 4 of the Open Systems Interconnection (OSI) model. This refers to the single point of contact for clients. Further, it helps in distributing the inbound flows that appear at the load

balancer's front end to backend pool instances. These flows are as per the configured load-balancing rules and health research. However, the backend pool instances can be Azure Virtual Machines or instances in a virtual machine scale set.

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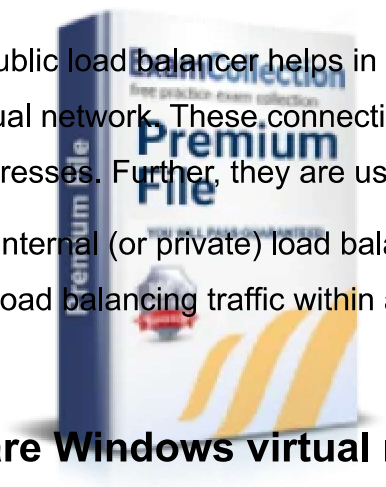
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Explain the public and private load balancer.



- A public load balancer helps in providing outbound connections for virtual machines (VMs) within a virtual network. These connections are achieve by translating private IP addresses to public IP addresses. Further, they are used for load balancing internet traffic to your VMs.
- An internal (or private) load balancer is use where private IPs are needed at the frontend only. They are for load balancing traffic within a virtual network.

What are Windows virtual machines in Azure?

Azure Virtual Machines (VM) or Windows Virtual Machines refers to an on-demand, scalable computing resource that Azure provides. VM helps in taking over the control of the computing environment. Moreover, the Azure VM provides the flexibility of virtualization without having any need for buying and maintaining the physical hardware running it. But, there is a need for maintaining the VM during performing tasks like configuring, patching, and installing the software running it.

I want to create a VM. What things should I consider before creating a VM?

There is always a multitude of design considerations while creating an application infrastructure in Azure. However, before starting, take a look at the following aspects of a VM:

- Firstly, the names of your application resources
- Secondly, the location where the resources are store
- Thirdly, the size of the VM
- Then, the maximum number of VMs that can be built
- After that, the operating system that the VM runs
- Next, the configuration of the VM after it starts
- Lastly, the related resources that the VM requires

What is the main role of the Azure Service Level Agreement (SLA)?

Azure SLA service makes sure that while sending two or more role instances for each role, access to your cloud service will be maintained 9 out of 10 times. This explains Microsoft's commitments for uptime and connectivity.

What do you understand by Azure Service Bus?

Azure Service Bus can be defined as a cloud technology use for messaging and communicating between different applications and devices. This helios the message brokers for conducting the processing of messages and messaging stores for caching the messages. Queue and topic are the entities in Azure Service Bus.

What is the role of the hybrid cloud in Azure?

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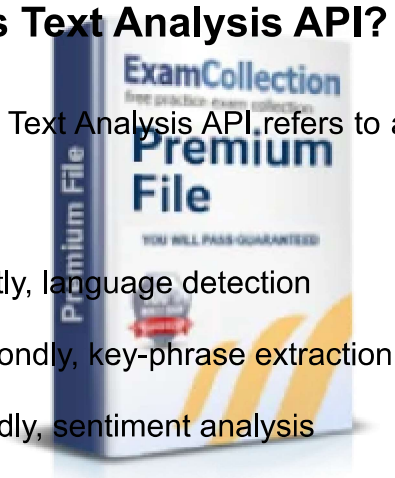
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Hybrid clouds refer to the combination of public and private clouds bounded together by technology. However, by allowing data and applications for moving between private and public clouds, a hybrid cloud gives your business greater flexibility, more deployment options, and helps in optimizing your existing infrastructure, security, and compliance.

What is Text Analysis API?



Azure ML Text Analysis API refers to a cloud-based service used for the NLP of raw Text. This performs four tasks:

- Firstly, language detection
- Secondly, key-phrase extraction
- Thirdly, sentiment analysis
- Lastly, entity recognition.

What is the major role of the Azure Web App?

Azure Web App provides high scalability, Multi-Language support, DevOps Optimization, Compliance and Security, Easy Integration with Visual Studio and Code, Serverless Code, and low maintenance cost.

What is the role of the dead letter queue in Azure?

The role of the dead-letter queue is to hold messages that can’t be deliver to any receiver, or messages that can no longer be processed. After this, messages can be remove from the DLQ and inspected. Using the help of an operator an application might correct issues and resubmit the message and log the fact that there was an error. However, the DLQ is mostly similar to any other queue, except that messages can only be submitted via the dead-letter operation of the parent entity.

Explain the term Verbose Monitoring in Azure.

It is use for collecting data performance matrix inside the particular role instance for analyzing the circumstances that appear while processing the application.

Is there any support for continuous integration/deployment of custom containers in Azure?

Yes, for private registries, you can update the container by stopping and then re-starting your web app. Moreover, you can also modify or add a dummy application setting for forcing an update of your container.

Name the types of RBAC controls in Microsoft Azure.

The types of RBAC controls are:

- Firstly, the Owner. This is for providing complete access to all resources including the right for assigning access to others.
- Secondly, Contributor. This helps in building and managing all types of Azure resources but it cannot provide access to others.
- Lastly, Reader. Using this, you can view existing Azure resources.

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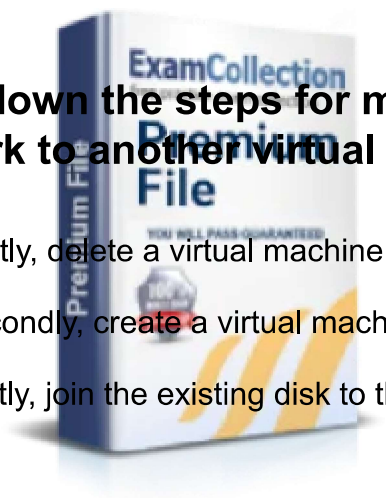
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I am facing an Azure Virtual Machine encounters issues generated by user configurations or host infrastructure. What should I do?

For this kind of issue, move the virtual machine to a different host. Take help of redeploying blade virtual machine for moving it.



Write down the steps for moving an Azure Virtual Machine from one virtual network to another virtual network?

- Firstly, delete a virtual machine in VNET1
- Secondly, create a virtual machine in VNET2
- Lastly, join the existing disk to the newly created VM

What is the process of resizing a virtual machine in the Azure Availability Set?

- Firstly, terminate all VMs in the availability set
- Secondly, resize the one VM
- Thirdly, begin the resizing of the VM that you want
- Lastly, after successfully resizing, start with the other VMs

Company ABC provides manufacturing facilities globally. Each facility consists of various machines that produce products. The machines create many messages daily for reporting progress, quality control metrics, and alerts. I want to design a solution for receiving and processing messages from the machines. Which Azure service will best suitable for this?

For this, I will use Azure Event Hubs. This service refers to a highly scalable data streaming platform and ingestion service which has the ability to receive and operate millions of events per second. So, this process and stores events, data, or measures created by distributed software and devices. Further, the data sent can be convert and store using any real-time analytics provider.

What is the role of Azure Diagnostics API?

Azure Diagnostics API is use for collecting diagnostic data like performance monitoring, and system event log from the applications that are running on Azure. Further, it can be used for:

- Firstly, monitoring of the data
- Secondly, building visual chart representations
- Thirdly, creating performance metric alerts.

What do you understand by swap deployments?

For promoting a deployment in the Azure staging environment to the production environment, you can swap the deployments by moving the VIPs by which the two deployments are accessed. After deploying, the DNS name for the cloud service points to the deployment which is present in the staging environment.

Which class should I use while retrieving the data?

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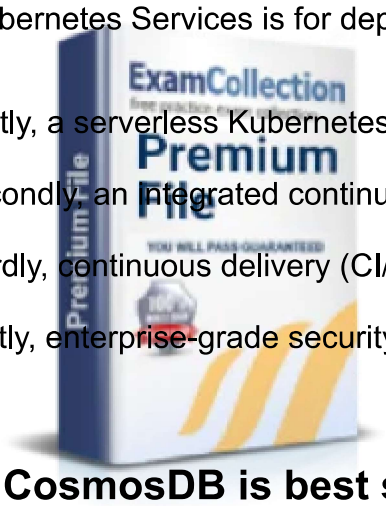
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The SPSite Data query is use for retrieving the data present in different lists. This is to sort and aggregates data using the help of SharePoint.

What is Azure Kubernetes Service (AKS)?

Azure Kubernetes Services is for deploying and managing containerized applications easily. This provides:



- Firstly, a serverless Kubernetes
- Secondly, an integrated continuous integration
- Thirdly, continuous delivery (CI/CD) experience
- Lastly, enterprise-grade security and governance.

Which CosmosDB is best suitable for providing temporary access to Cosmos DB to your application?

For getting temporary access to your Azure Cosmos DB account, you can use the read-write and read access URLs.

What are read-write and read access URLs in CosmosDB?

- Read-Write can be define as when you share the Read-Write URL with other users. This allows them to view and change the databases, collections, queries, and other resources linked with that specific account.
- Read can be define as when you share the read-only URL with other users. This allows them to view the databases, collections, queries, and other resources lined with that specific account. For example, if you want to share the output of a query with your teammates. So, you can provide them access by giving this URL.

Explain Virtual Machine scale sets in Azure.

VM scale sets refer to the Azure compute resource whose function is to deploy and manage a set of identical VMs. These scale sets provide a simple process for creating large-scale services targeting big compute, big data, and containerized workloads if all the VMs configured the same.

Which service should I use for achieving high availability by autoscaling to create thousands of VMs in minutes?

Virtual Machine Scale Sets can be used. This helps in creating large-scale services for batch, big data, and container workloads. Further, you can create and manage a group of heterogeneous load-balanced virtual machines (VMs). Moreover, here you can increase or decrease the number of VMs automatically in response to demand or depending on a schedule you define. This also helps in centrally managing, configuring, and updating thousands of VMs and provides higher availability and security for your applications.

How can we deploy Azure virtual machines on a physical server that can only be used by your organization?

For this, you can use Azure Dedicated Host. This offers physical servers that host one or more Azure virtual machines. Using this, your server is dedicated only to your organization and workloads with no involvement of other customers. This host-level isolation further helps in addressing the compliance requirements. Lastly,

after provisioning the host, you gain visibility and control over the server infrastructure and then, you can regulate the host's maintenance policies.

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Differentiate Azure SQL Database and SQL managed instance.

- Azure SQL Database refers to a fully managed platform as a service (PaaS) database engine that controls most of the database management functions like upgrading, patching, backups, and monitoring without user involvement. This always runs on the latest stable version of the SQL Server database engine. Moreover, it consists of PaaS capabilities that help in focusing on the domain-specific database administration and optimization activities that are critical for your business.
- Azure SQL Managed Instance refers to an intelligent, scalable cloud database service that joins the broadest SQL Server database engine compatibility with all the benefits of a fully managed platform as a service. This is compatible with the latest SQL Server database engine, providing a native virtual network (VNet) implementation that addresses common security concerns, and a business model favorable for existing SQL Server customers. Further, it allows existing SQL Server customers to lift and shift their on-premises applications to the cloud with minimal application and database changes.

What is the role of Azure Synapse Analytics?

Azure Synapse Analytics refers to an analytics service that is used for bringing together enterprise data warehousing and Big Data analytics.

What makes Azure Data lake storage different from Azure blob storage?

Blob storage good at non-text-based files that includes database backups, photos, videos, and audio files. Whereas data lake is designed for large volumes of text data. However, for using text file data to be loaded into my data warehouse, Data lake would be a better option.

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Explain the best practice of using dynamic variables for build pipelines in Azure DevOps?

This can be performed by associating a Variable Group with the build pipelines. However, variable groups are used for storing pipeline-based variables and can be associated with Azure Key Vault.

What do you recommend, if you are in the security administrator role for the company's Azure account. And you have to analyze security recommendations for multiple subscriptions and require to enforce strict compliance for them.

Firstly, create an initiative with built-in and custom policies for recommendations and allocate the initiative at the management group scope. However, for creating a compliance mechanism for multiple subscriptions, you should build an initiative and allocate it to a management group for good management.

Name the Application Gateway features that provide Web App protection from common exploits.

For this, you can use the Web application firewall feature of the application gateway.

Write down the Azure CLI command for creating a new Azure AD user.

The command is, *az ad user create.*

Write down the PowerShell cmdlet for encrypting a managed disk in Azure.

The answer is, *Set-AzVMDiskEncryptionExtension.*

You want to improve the Dockerfile with great readability and maintenance. You decided to use Multiple Stage Builds. What things will you consider while using Multiple Stage Builds?

- Firstly I will check for adopting Container Modularity
- Secondly, I will avoid including Application Data, any unnecessary packages, and then, select an Appropriate Base.

However, Multi-stage build is a new feature that needs Docker 17.05 or higher on the daemon and client. These are useful to anyone who has struggled to improve Dockerfiles while keeping them easy to read and maintain.

ABC company is using Azure DevOps for the build pipelines and deployment pipelines of Java-based projects. They need a technique for managing technical debt. What would you recommend?

Firstly, there must be configuring of the pre-deployment approvals in the deployment pipeline as analysis should be at the pre-deployment stage. Secondly, it integrates Azure DevOps and SonarQube. SonarQube is used for examining the technical debt.

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Explain the process of scaling up and down ins scalability.

- Scaling Up refers to adding more resources to the existing nodes. For example, adding more storage, or processing power.
- Scaling Out refers to adding more nodes to support more users.

However, any methods can be used for scaling up/out as applicable. However, the cost of adding resources depends on the volume change.

Explain lower latency interaction.

Low latency can be defined as the very little delay between the request time and the response time. However, it is applied to WebSockets. This means the data can be sent faster because of the established connection. Further, there is no need for extra packet roundtrips to create the TCP connection.

What is the role of Clustering?

Clustering is necessary for achieving high availability for server software. This helps in reaching the availability or zero downtime in service. Further, by building a cluster of more than one machine, you can reduce the chances of our service going un-available in case one of the machines fails.

What is the ACID property?

ACID property refers to basic rules that have to be satisfied by every transaction for preserving integrity. There are properties and rules which include:

1. Atomicity

It’s an all or none concept which helps in enabling the user to be assured of handling the incomplete transactions. In this, every transaction is taken as one unit and either run to completion or is not executed at all.

2. Consistency

This property defines the uniformity of the data. However, it implies that the database remains consistent before and after the transaction.

3. Isolation

This property defines the number of the transaction executed concurrently without leading to the inconsistency of the database state.

4. Durability

This property makes sure after the transaction is committed, it will be stored in the non-volatile memory. And, then even system crash cannot affect it anymore.

Explain CAP Theorem.

The CAP Theorem states that it is impossible to create an implementation of read-write storage/system in an asynchronous network that satisfies the following properties:

- Firstly, Availability

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- Secondly, Consistency
- Lastly, Partition tolerance

Explain the microservice approach and monolithic app.

- Microservice architecture refers to a form of the service-oriented architecture structure. This arranges an application as a collection of loosely coupled services. In this, the services are fine-grained and the protocols are lightweight.
- A monolithic application refers to a single-tiered software application that allows the user interface and data access code to merge into one program from one platform. However, this is self-contained and independent from other computing applications.

Name the tools used by an IT Solutions Architect.

Some of the tools include:

- Firstly, **Nagios**. This refers to an open-source application used for monitoring networks, systems, and infrastructure.
- Secondly, **Git.** This refers to a version control system used for tracking the changes made in source codes during the development of software.
- Thirdly, **Travis.** This is an integrated tool used for creating and testing software projects.
- Then, **Java**. This is an object-oriented coding language for developing applications.
- Lastly, **Docker.** This provides an application containerization platform for packaging software or applications in filesystems.

What is the process of improving the existing software?

You can perform an upgrade to improve an existing system. There always updates in the software, so it is important to keep it up-to-date for getting a smooth performance and keeping it secure.

Explain the metrics for validating solution compliance with enterprise architecture.

Testing is use for validating a software solution and ascertain its compliance with enterprise architecture. This discloses the compatibility of the application with the existing architecture. Further, you can also use architecture documentation for determining solution compliance.

Name the components of the web applications.

The components are:

- Firstly, the **View layer.** This provides an interface to the application for receiving information in and out of the application.
- Secondly, the **Business layer.** This receives user requests from the internet, processes them, and decides the routes using which the information will be accessed.
- Thirdly, the **Data access layer**. This keeps the code that clients use for pulling information from their data stores like flat files, databases, or several web services.
- Lastly, the **Error security, handling, and logging.** This handles the errors to make users feel secured and informed.

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Which service in Azure can be used to manage resources?

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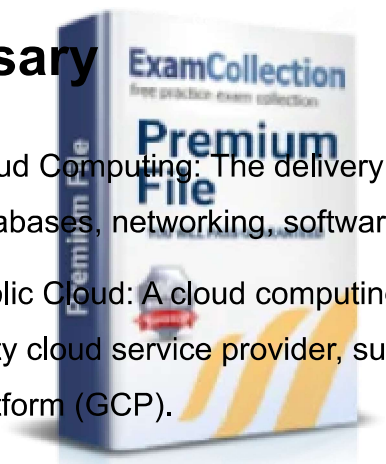
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Azure Resource Manager manages the resources in Microsoft Azure. It uses a simple JSON script for deploying, managing, and deleting all the resources together.

Glossary



- Cloud Computing: The delivery of computing resources over the internet, including servers, storage, databases, networking, software, analytics, and intelligence.
- Public Cloud: A cloud computing environment where resources are owned and operated by a third-party cloud service provider, such as Microsoft Azure, Amazon Web Services (AWS), or Google Cloud Platform (GCP).
- Private Cloud: A cloud computing environment where resources are owned and operated by a single organization for its own use.
- Infrastructure as a Service (IaaS): A cloud computing service model where a third-party cloud provider offers virtualized computing resources, such as virtual machines (VMs), storage, and networking, to customers.
- Platform as a Service (PaaS): A cloud computing service model where a third-party cloud provider offers a platform for customers to develop, run, and manage applications, without needing to manage the underlying infrastructure.
- Azure Active Directory (Azure AD): A cloud-based identity and access management service that provides secure access to Azure resources and other Microsoft services.
- Azure DevOps: A cloud-based service that provides development tools, services, and support for continuous integration and delivery (CI/CD) pipelines.
- Learn Azure Functions: A serverless computing service that enables developers to run event-driven code in response to events or triggers.
- Azure Kubernetes Service (AKS): A managed Kubernetes service that simplifies the deployment and management of containerized applications.
- Azure Monitor: A cloud-based monitoring and management service that provides visibility into the health and performance of Azure resources and applications.
- Learn Azure Resource Manager: A service that enables developers to provision and manage resources in Azure using templates and declarative syntax.
- Azure Security Center: A cloud-based service that provides unified security management and advanced threat protection for Azure resources.
- Azure Site Recovery: A disaster recovery service that replicates and fails over applications and workloads to a secondary site in the event of an outage or disaster.

Final Words

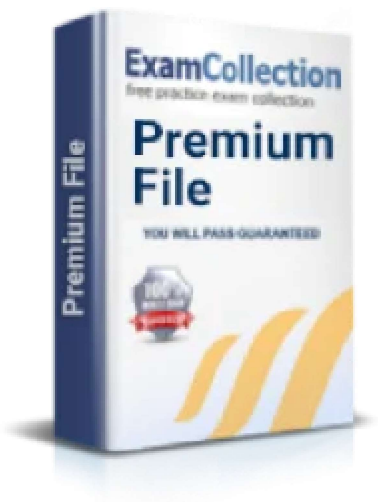
Above, we have discussed the top Microsoft Solution Architect interview questions. With the rapid growth of the cloud sector, the role of Microsoft solution architect is getting a lot of importance in the job sector. As they are responsible for handling various tasks like designing and implementing solutions on Microsoft Azure and managing services like compute, network, storage, and security. So, if you have the required knowledge and skills then, start preparing to earn this role. Take help from the questions provided in this blog and ask for any doubt in the comment sections.

Become a certified solution architect by passing the AZ-303 and AZ-304 Exam!

Pulkit Dheer **ExamCollection**

With a background in Engineering and a great enthusiasm for writing, Pulkit focuses on intensive research to create targeted content. He brings his years of learning and experience to his current role. With a zeal towards technological research and powerful use of words dedicated to inspire and help professionals onset their career.

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